



793D Converter Removal & Installation Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Currently the tools for removal and installation of the 793D OHT Torque Converter are chains and eyebolts (Fig. 1). These tools were designed for high safety, but we felt we have improved upon the process.

This document shows a tool (preliminary design) for the converter removal and installation more safely and quickly.

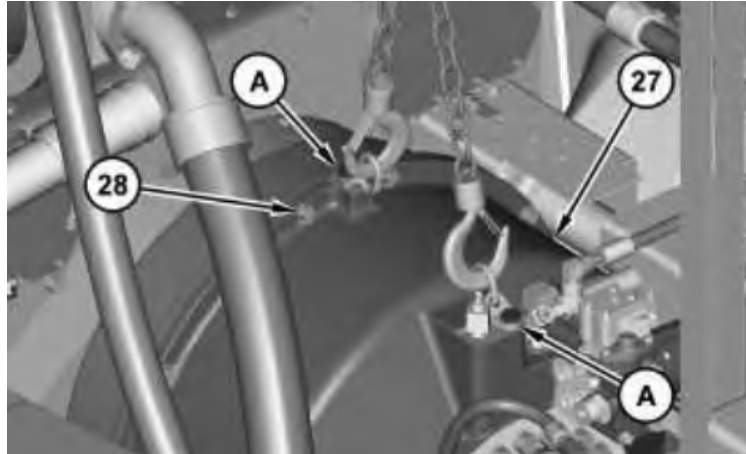


Figure 1. SIS procedure for the Converter removal

2.0 Best Practice Description

The tool consists of three bent and welded plates as shown in Figure 2.

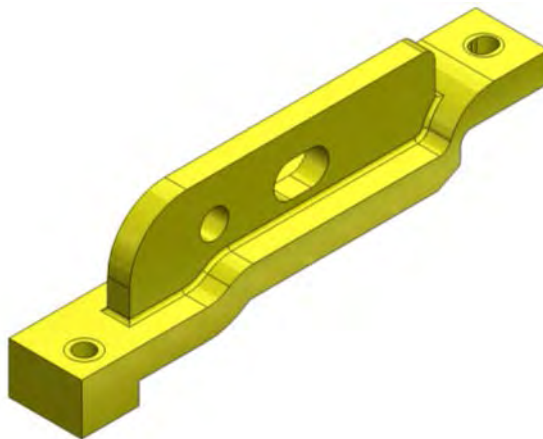


Figure 2. Converter Removal & Installation Tool

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The two holes that are observed, the slot type and the circular one have specific functions. In the first one, the shackle for lifting during removal is located; it has this shape so the component can be inclined and it allows a better coupling. The other circular hole is used for a guide rope while the component is being lifted. See figure 3.

3.0 Implementation Steps

Steps for Removal

1. Refer to document *REN8328-05 Torque converter Removal*.
2. Clean the bolt holes on the converter top
3. Attach the removal tool (1) to torque converter (6) using 5/8 – 11 x 2.50 UNC Bolt (2), 5/8 – 11 x 3.25 UNC (3), and two 5/8 washers (4). Tighten bolts to 60 N·m (see figure 4)
4. Attach the screw pin shackle (5) to straight slot in R&I tool
5. Attach appropriate overhead lifting device of minimum 2 ton capacity to raise and support the tool with the converter (The weight of torque converter is approximately 1043 kg).

Steps for Install

1. Clean the bolt holes on the converter top
2. Attach the removal tool (1) to torque converter (6) using 5/8 – 11 x 2.50 UNC Bolt (2), 5/8 – 11 x 3.25 UNC (3), and two 5/8 washers (4). Tighten bolts to 60 N·m (see figure 4)
3. Attach Screw Pin Shackle (5) to circular hole in R&I tool
4. Attach appropriate overhead lifting device to raise and support the tool with the convert.
5. Continue with the guide *REN8328-05 Torque converter Install* for install the other components



Figure 3. Tool position on the top converter

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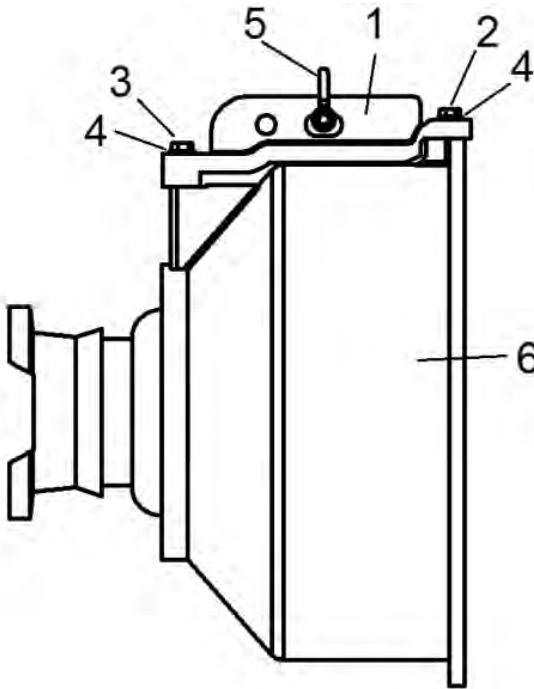


Figure 4. (1) Tool, (2) 0S-1625 bolt, (3) 1D-4585 Bolt, (4) 8T-4122 Washers, (5) 5P-8622 Shackle, (6) Torque converter (removal position)

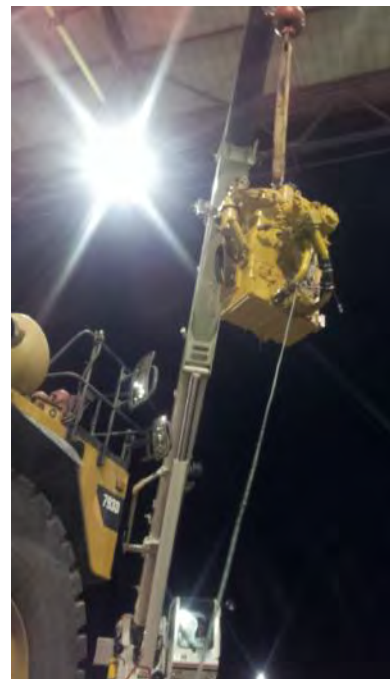


Figure 5. Torque converter during removal

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The tool can be used in the following converter:

Table 1. Converter that apply	
Part Number	Description
240-7186	Torque Converter Group

4.0 Benefits

With this tool it is easier to control the inclination and alignment of the coupling to fly wheel housing. This also helps keep things more level.

5.0 Resources Required

- Hardware list (see table below)
- Technical Personnel.
- Material and equipment required to make the tool.

Table 2. Hardware Required		
Part Number	Description	Qty
0S-1625	5/8-11 x 2.50 inch Bolt	1
1D-4585	5/8-11 x 3.25 inch Bolt	1
8T-4122	5/8 inch Washer	2
5P-8622	Screw Pin Shackle (6.5 Ton)	1

6.0 Supporting Attachments / References

- SIS document: SENR 8328-05 (Torque Converter Remove – SMCS: 3101-011)
- SIS document: SENR 8328-05 (Torque Converter Install – SMCS: 3101-012)

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- 793D Converter R&I Tool Drawing.

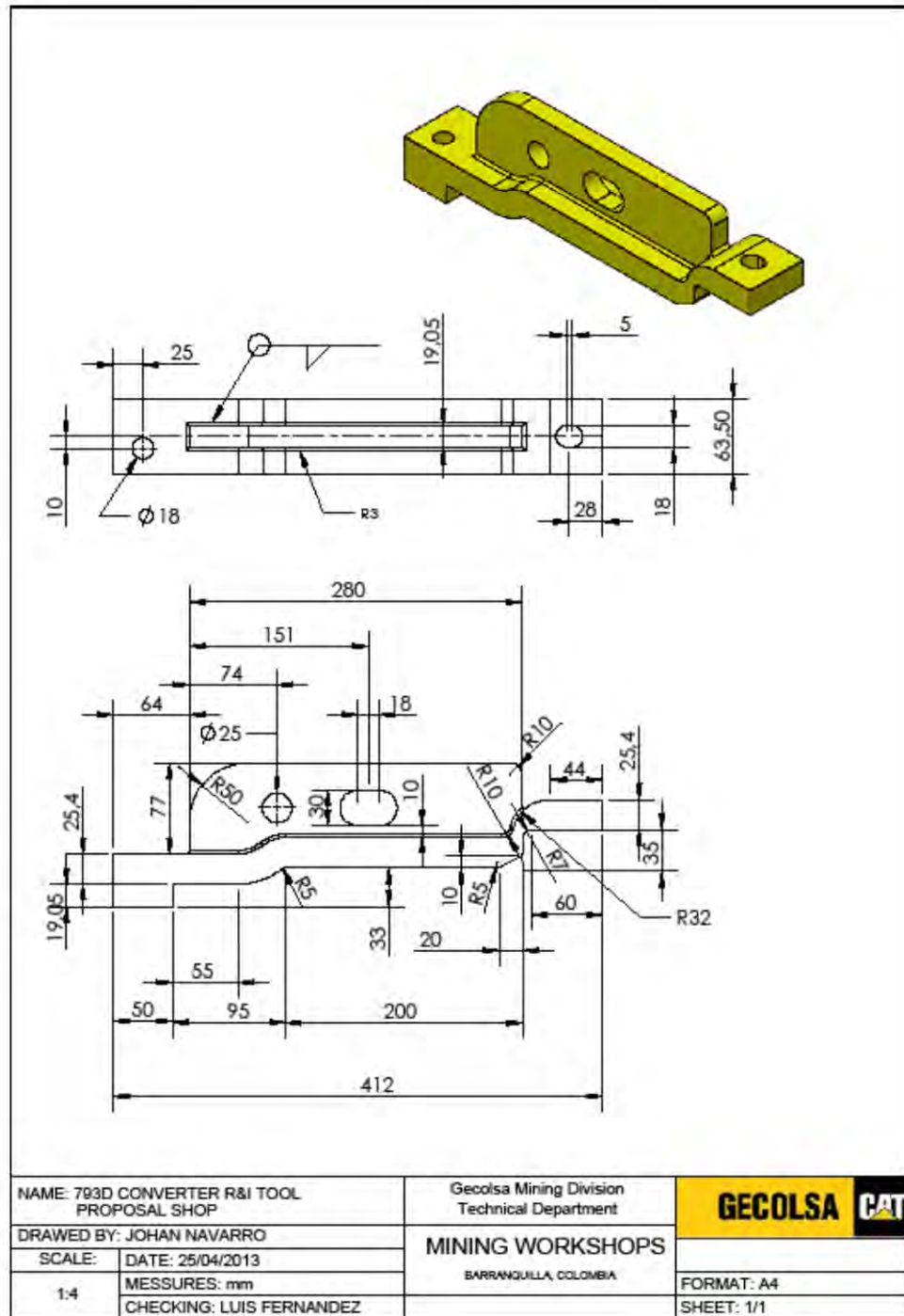


Figure 6. Tool Drawing

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- Finite Element Analysis for R&I Tool

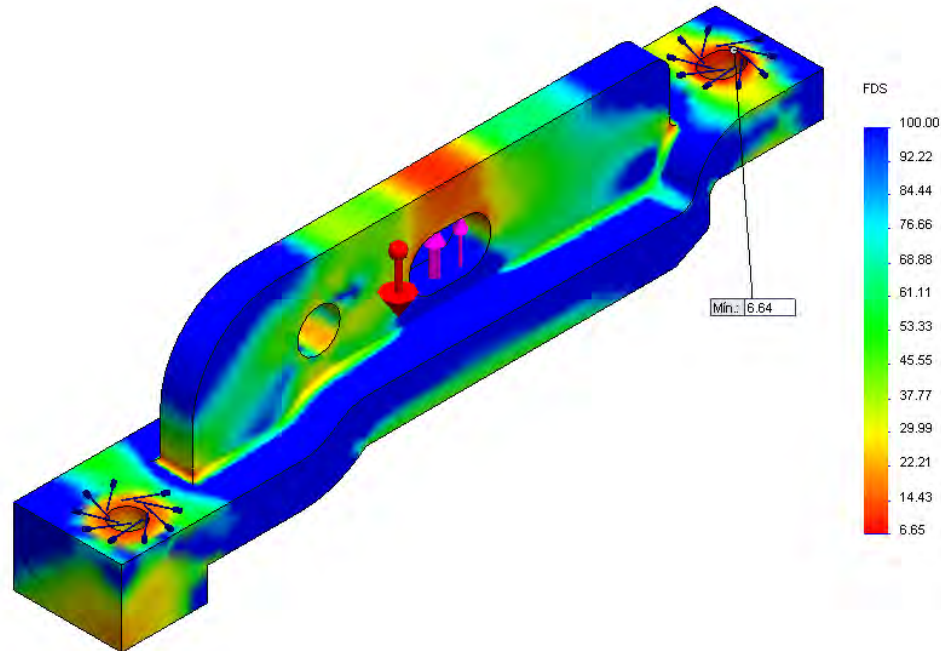


Figure 7. Safety Factor Analysis

Physical Properties		
Material	AISI 1045 Steel, cold drawn	
Mass	7.982 Kg	
Density	7850 Kg/m ³	
Yield Strength	530 MPa	
Tensile Strength	625 MPa	
Shear Modulus	80 Gpa	
Study Properties		
Analysis type	Static	
Mesh type	Solid mesh	
Mesh Quality	Second order tetrahedral elements	
Model type	Lineal elastic isotropic	
Failure criteria	Von Mises Stress Maximum	
Fixtures		
Type	Fixtures details	
Foundation Bolt	Quantity	2
	Nominal Diameter	5/8 inch
	Pre-load (Torque)	60 N·m
Virtual Wall	Entities	2 faces (below faces)
	Wall type	Bonded
	Friction coefficient	0.15
Loads		
Gravity	9.81 m/s ²	
Vertical Force	10300 N (400 kg approx.)	
Study Results		
Von Mises Maximum Stress	79.9 Mpa	
Minimum Safety Factor	6.64	

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Based on the finite element analysis and considering the minimum safety factor given by the ASME B30.20-2003 standard: *Below-the-Hook lifting device*, the tool loading capacity is 2200 Kg

Editor's Note: Caterpillar will soon have a tool similar to this available to Dealers. You may request plans from Gecolsa to fabricate your own or order one from Caterpillar when it becomes available. The use of the this installation tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

7.0 Related Best Practices

8.0 Acknowledgements

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